Memory Circuits and System Design

Project 3 - DRAM Engineering Lab

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Design and simulate the core read-path of a DRAM array consisting of three key building blocks

1T-1C DRAM Cell

- The 1T-1C DRAM cell consists of one access transistor and one storage capacitor. The data is stored in the form of charge on the capacitor. A logical '1' is represented by a charged capacitor (approximately equal to V_{DD}), while a logical '0' corresponds to a discharged capacitor.
- During the read operation, the bitline (BL) is precharged to $V_{DD}/2$ (0.55V in this case). When the wordline (WL) is asserted, the access transistor turns ON and connects the storage node to the bitline. This results in charge sharing between the storage capacitor and the bitline capacitance.

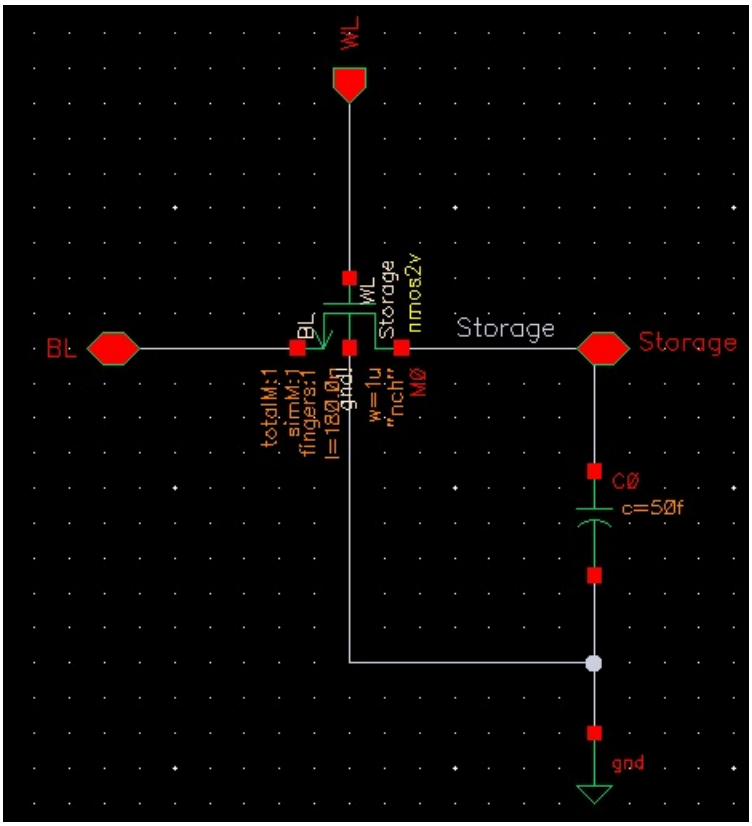


Figure 1: DRAM Cell Schematic

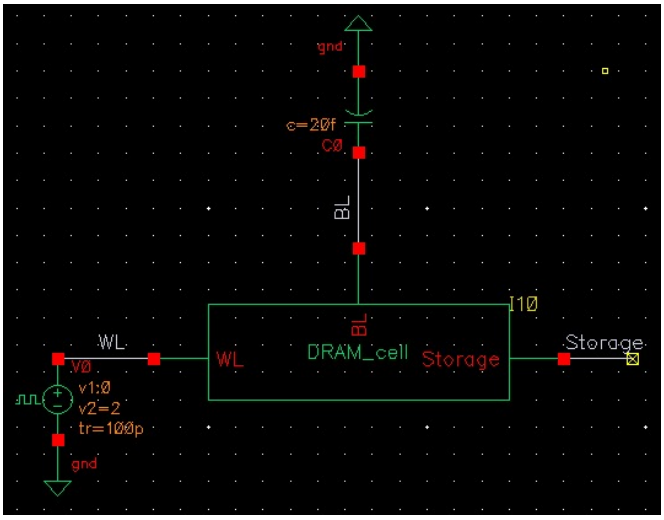


Figure 2: DRAM Cell Testbench Schematic

The DRAM cell is simulated using a precharged bitline (BL) initialized to 0.55V which is connected to a capacitive load of 20 fF. The storage node is initialized to 1.1V, representing a stored logic '1'. The wordline (WL) is driven using a pulse source with a boosted voltage of 2V to emulate wordline boosting (VPP).

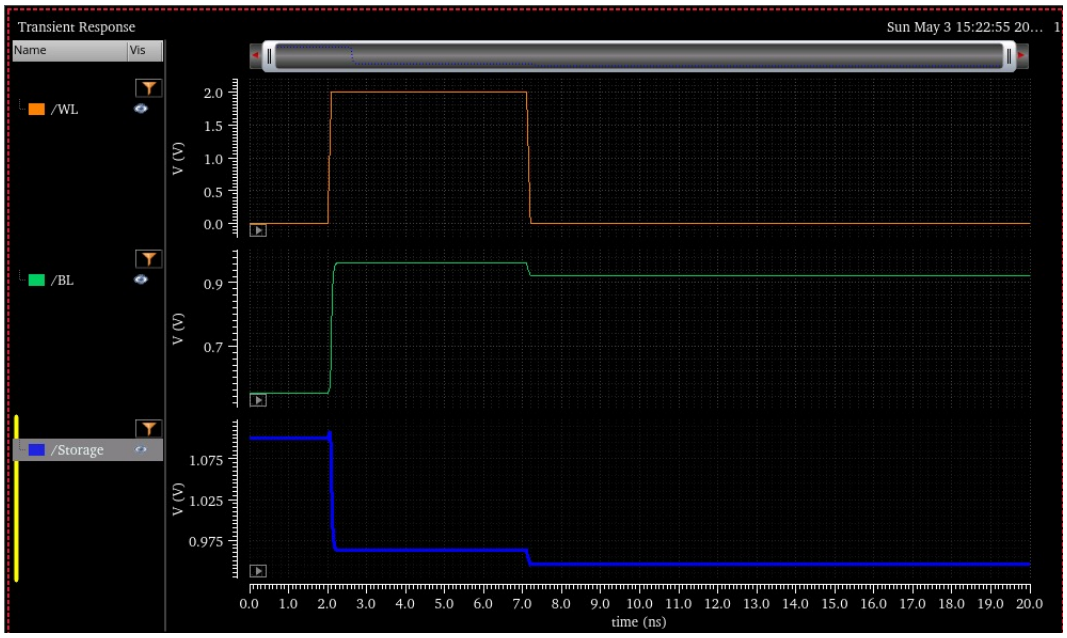


Figure 3: DRAM Cell Simulation

- Initially, the bitline is at approximately 0.55V (half of V_{DD}), and the storage node is at 1.1V. When the wordline is asserted at around 2ns, the access transistor turns ON, allowing charge sharing between the storage capacitor and the bitline.
- As a result, the bitline voltage increases from 0.55V to approximately 0.95V, while the storage node voltage decreases from 1.1V to around 0.97V. This confirms the transfer of charge from the storage node to the bitline.
- When the wordline is turned OFF, both nodes settle to their respective voltages, indicating that the read operation has partially discharged the storage node meaning the read operation is destructive, and the original data must be restored using a sense amplifier in a complete DRAM system..

Sense Amplifier

- The sense amplifier is a cross-coupled latch structure used to amplify the small voltage difference generated on the bitlines during the DRAM read operation.
- Initially, both BL and BLB are precharged to approximately $V_{DD}/2$. A small differential voltage is introduced due to charge sharing in the DRAM cell.
- When the Sense Enable (SE) signal is asserted, the cross-coupled inverters form a positive feedback loop. This causes one node to be pulled towards V_{DD} while the other is driven towards ground.

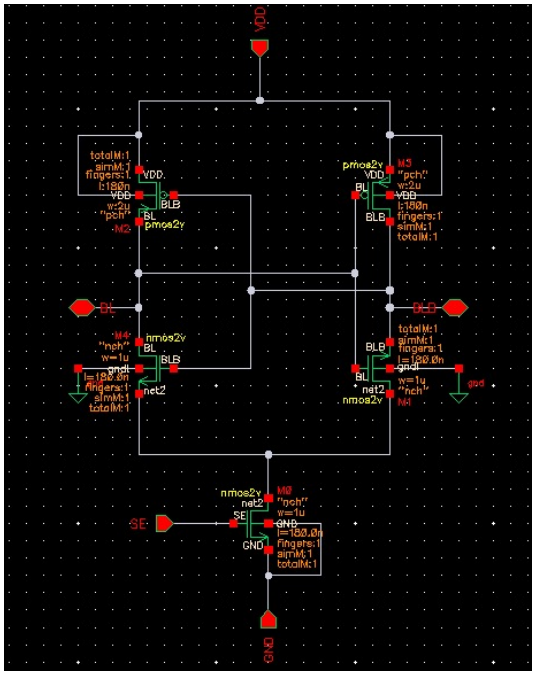


Figure 4: Sense Amplifier Schematic

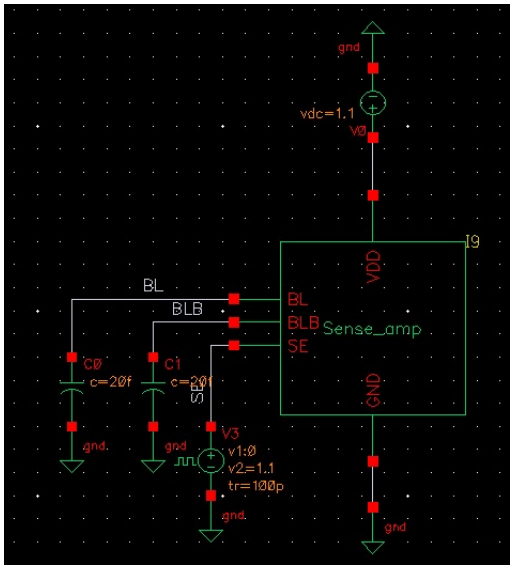


Figure 5: Sense Amplifier Testbench Schematic

The sense amplifier is tested by initializing BL to 0.56V and BLB to 0.54V which are slightly above and below $V_{DD}/2$. Capacitive loads of 20 fF are connected to both BL and BLB to model the bitline capacitance. The sense amplifier is powered with $V_{DD} = 1.1V$, and a pulse signal is applied to the sense enable (SE) input to activate the amplification process.

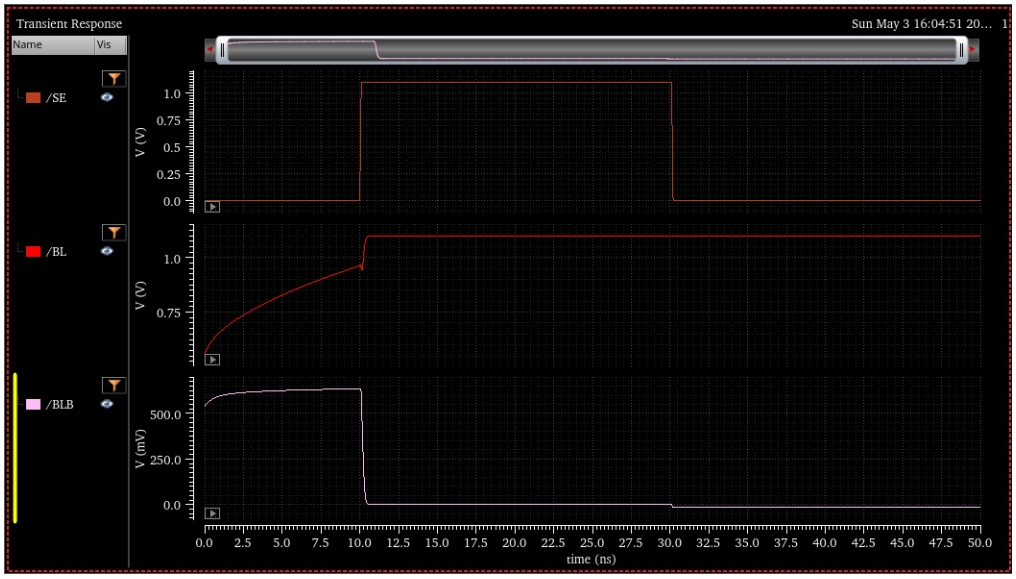


Figure 6: Sense Amplifier Simulation

- Initially, both BL and BLB are close to 0.55V, with BL being slightly higher than BLB. Before the sense enable (SE) signal is activated, both nodes remain near their initial values with a gradual drift due to capacitive effects and absence of an equalization circuit.
- When the SE signal is asserted at around 10ns, the sense amplifier is enabled, and the cross-coupled structure begins amplifying the small voltage difference. As a result, BL is driven rapidly towards V_{DD} (approximately 1.1V), while BLB is pulled down towards 0V demonstrating the regenerative action of the sense amplifier.
- After amplification, both nodes settle into stable logic levels, indicating successful latching of the sensed data.

2-to-4 Row Decoder

- The 2-to-4 row decoder is used to select one wordline out of four possible rows based on two input address bits.
- The decoder generates four outputs (WL0, WL1, WL2, WL3), where only one output is active (high) at a time depending on the input combination. This ensures that only one row of the DRAM array is accessed during a read or write operation.

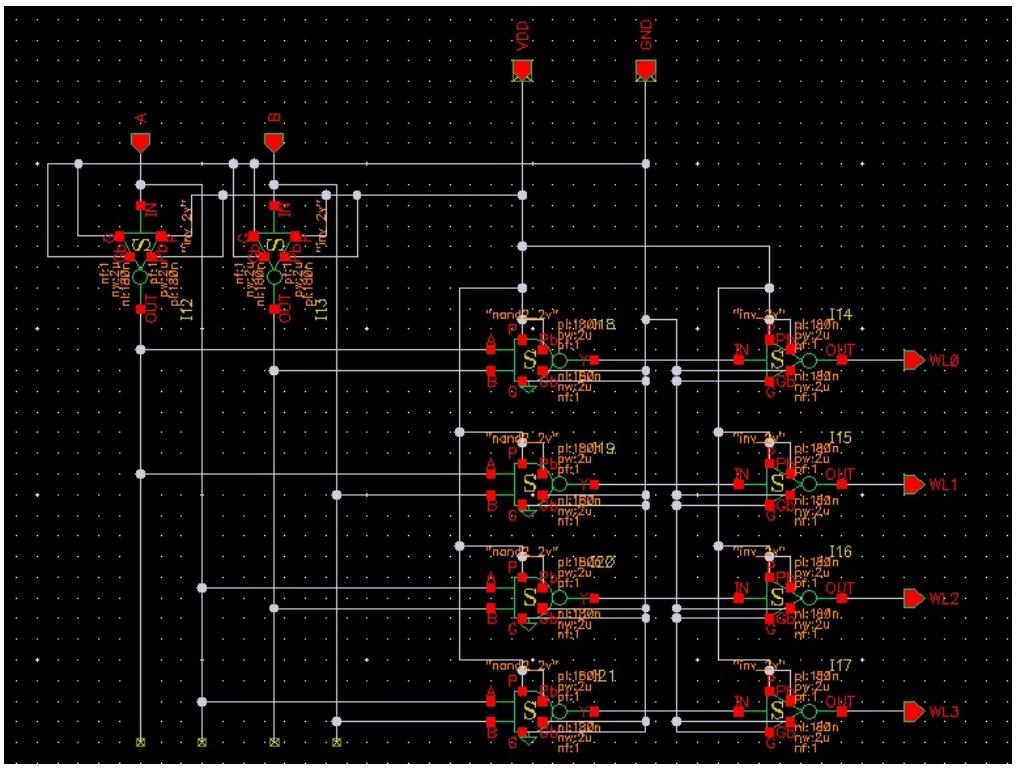


Figure 7: 2-to-4 Row Decoder Schematic

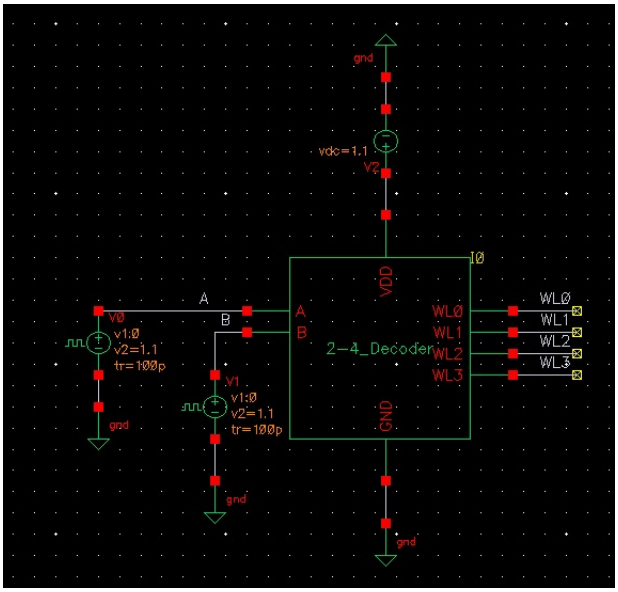


Figure 8: 2-to-4 Row Decoder Testbench Schematic

The 2-to-4 decoder is tested using two input signals (A and B) generated using pulse sources. The inputs are driven with different frequencies to ensure that all possible input combinations (00, 01, 10, 11) are exercised during the simulation.

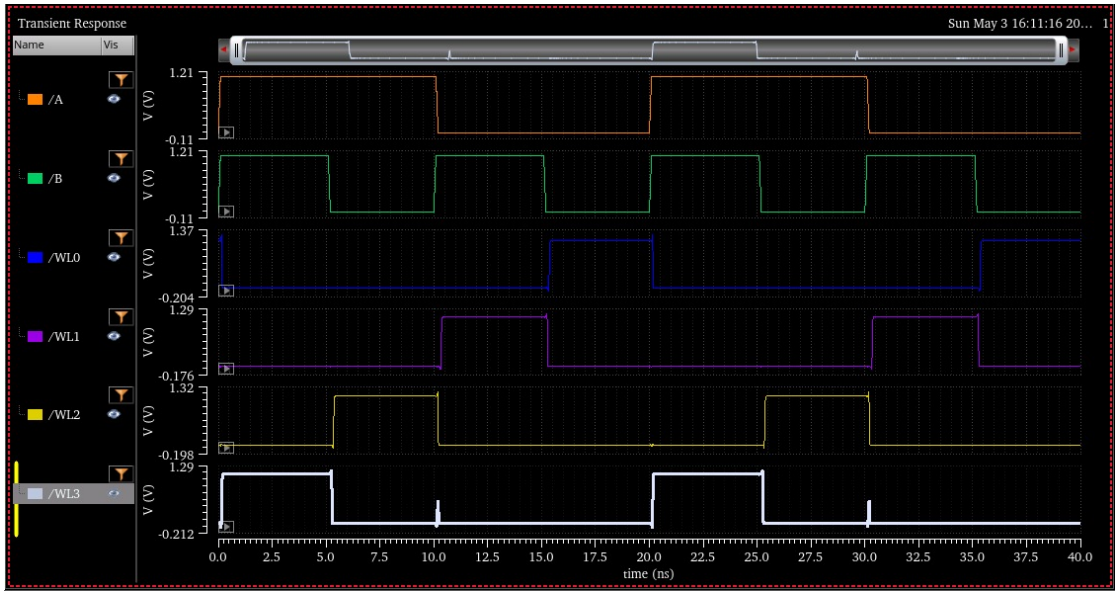


Figure 9: 2-to-4 Row Decoder Simulation

- For each input combination, exactly one of the outputs (WL0–WL3) is asserted high, while the remaining outputs stay low.

  Specifically:

  - WL0 is high when A = 0 and B = 0
  - WL1 is high when A = 0 and B = 1
  - WL2 is high when A = 1 and B = 0
  - WL3 is high when A = 1 and B = 1

- The outputs reach near V_{DD} for logic high and near 0V for logic low, confirming proper logic-level operation.

Final Block

In this section, the 1T-1C DRAM cell is integrated with the sense amplifier to demonstrate the core read path operation. This implementation is carried out without using a level shifter. Instead, the wordline (WL) is driven using a piecewise linear (PWL) voltage source to emulate the boosted wordline voltage ($V_{PP} \approx 2.0V$).
The WL input is defined as follows:

Time	Voltage
0	0
2ns	0
2.1ns	2.0V
30ns	2.0V
30.1ns	0
50ns	0

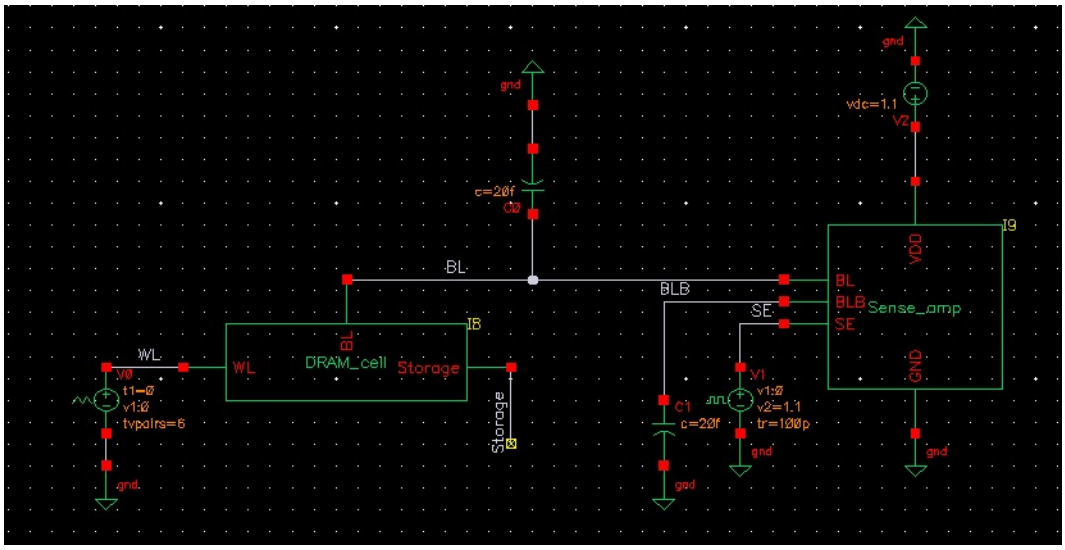


Figure 10: Final Integration Schematic

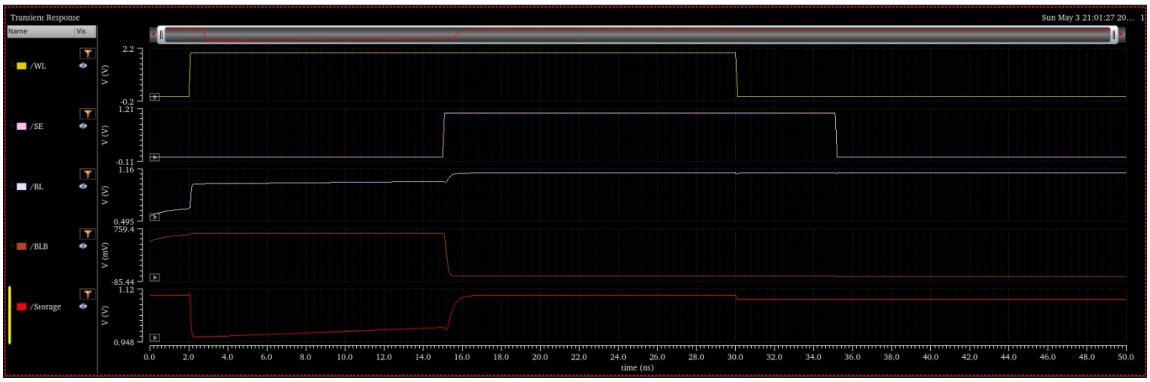


Figure 11: Final Simulation

- Initially, both BL and BLB are equal to the precharged voltage 0.55V. When the wordline is asserted at around 2.1ns, the access transistor turns ON, enabling charge sharing between the storage node and the bitline.
- As a result, the bitline voltage increases slightly, while the storage node voltage decreases, indicating the transfer of charge. During this phase, a small differential voltage is developed between BL and BLB.
- When the sense enable (SE) signal is asserted at around 15ns, the sense amplifier is activated. Due to the positive feedback mechanism, the small voltage difference is rapidly amplified, driving BL towards V_{DD} and BLB towards ground.
- Simultaneously, the storage node voltage is restored towards its original value, demonstrating the refresh operation that compensates for the destructive read. After the wordline is turned OFF at around 30ns, the cell is isolated, and the sensed data remains latched in the sense amplifier.